Galois Fields Worksheet

Names:

GF(2^8) math summary. Each bit of a byte is the coefficient of one power of x, so $x^7 + x^2 + 1$ is `0b10000101`.

- Addition (and subtraction) is XOR.
- Multiplication by x is a left shift. If the coefficient of x^8 is 0, you are done. Otherwise, reduce modulo the irreducible polynomial $m(x)$ by XORing with it.

Consider a Galois field GF(2^8) with characteristic polynomial $m(x) = x^8 + x^4 + x^3 + x + 1$.

1. Express the characteristic polynomial as a hexadecimal number.

2. Express $c = x^7 + x^6 + x^3 + x^2 + x + 1$ in hexadecimal.

3. Let $d = x^7 + x^3 + x$. Compute $e = c + d$.

4. Compute $-d$.

5. Compute $c \cdot x$.

6. Compute $c \cdot x^2$.